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Roll No. :.....

PYB-101
Examination –Dec- 2023
B.Tech. I/II Sem : Common to all Branches
Subject : Applied Physics

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	Wave group associated with a moving material particle travels with the same velocity. (i) True (ii) False	02	2	1
	(b)	A wave function ψ is normalized if (i) $\int \psi\,\psi^*\,d\tau = \infty$ (iii) $\int \psi\,\psi^*\,d\tau = 0$ (ii) $\int \psi\,\psi^*\,d\tau = 1$ (iv) $\int \psi\,\psi^*\,d\tau = N$	02	2	1
	(c)	Explain the term phase velocity and group velocity.	03	1	1
		OR			
		If the uncertainty in position of an election is 5×10^{-10}m, calculate the uncertainty in momentum.	03	3	1
	(d)	Derive an expression for Compton shift in the wavelength of a photon after scattering from an electron.	05	3	1
		OR			
		Derive Schrödinger time independent equation for matter wave.	05	3	1
Q.2	(a)	Lifetime of an atom in the excited state is usally (i) 10^{-2} sec (ii) 10^{-5} sec (iii) 10^{-8} sec (iv) 10^{-11} sec	02	2	2
	(b)	The refractive index of cladding is always .....than that of the core region.	02	1	2
	(c)	Explain the stimulated emission of radiation.	03	2	2
		OR			
		Find the numerical aperture of optical fiber having clad index 1.43 and core index of 1.54.	03	3	2
	(d)	Discuss the working of Ruby or He-Ne laser with energy level diagram.	05	3	2
		OR			
		Derive the expression for acceptance angle of the step index optical fiber.	05	2	2
Q.3	(a)	In an intrinsic semiconductor, at certain temperature, the product of concentration of electrons and holes is always constant. (i) True (ii) False	02	2	3

	(b)	If the applied forward voltage is V_f, and forward and reverse saturation current I_f and I_o respectively of p-n junction diode , then forward characteristic equation is (i) $I_f = I_0\left(e^{V_f/kT}+1\right)$ (iii) $I_f = I_0\left(e^{V_f/kT}-1\right)$ (ii) $I_f = I_0\left(e^{1/kT}-1\right)$ (iv) $I_f = I_0\left(e^{1/kT}+1\right)$	02	4	3
	(c)	Explain effective mass of electron in lattice.	03	2	3
		OR			
		Explain V-I characteristic of Solar cell.	03	2	3
	(d)	Discuss the direct and indirect band gap in E-k form of the band diagram, and Fermi energy levels in semiconductor.	05	3	3
		OR			
		Write LED structure and its characteristics.	05	2	3
Q.4	(a)	The temperature at which a normal materials turns into a superconductor is called as: (i) Normal temperature (ii) High temperature (iii) critical temperature (iv) None of these	02	2	4
	(b)	The nanomaterials can be synthesized in two ways are.....	02	1	4
	(c)	Explain the Type- I superconductor.	03	2	4
		OR			
		Write the dimension wise classification of nano materials	03	2	4
	(d)	Explain temperature dependence of critical field and Meissner effect of superconductor.	05	2	4
		OR			
		Discuss the application of nanotechnology in various fields.	05	3	4
Q.5	(a)	Which one is not correct for non-polar dielectric: (i) center of gravity coincide (ii) symmetric structure (iii) permanent dipole moment (iv) zero dipole moment	02	2	5
	(b)	The Piezoelectricity of certain crystals become electrically polarized when: (i) magnetized (ii) optically stimulated (iii) externally stressed (iv) None of these	02	2	5
	(c)	Explain Dielectric Polarization.	03	2	5
		OR			
		Explain direct and converse Piezoelectric effect.	03	2	5
	(d)	Describe Gauss law in Dielectric materials.	05	3	5
		OR			
		Discuss Piezoelectric materials and its use as sensor.	05	3	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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